SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA11

COURSE OUTCOME COVERED

CO2: Design UML diagrams to represent system requirements and architecture. *(BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 6 **Total Marks:30**

Annexure A

Scenario-Based

Code of Conduct:

I ___K.Divya(192511316)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Scenario-Based

1. Use Case Diagram Scenario

A **Smart Pharmacy Management System** enables customers to search medicines, upload prescriptions, place medicine orders, and make online payments. Pharmacists verify prescriptions and dispense medicines, while administrators manage medicine inventory and user accounts. The interactions among system users are not clearly defined.
Based on this scenario, analyze the system requirements, identify all actors and use cases, and design a **Use Case Diagram** showing appropriate relationships (**include, extend, and generalization**) to ensure complete system functionality.
(5 Marks)
Answer:
Analyze the system requirements:
The Smart Pharmacy Management System allows customers to search medicines, upload prescriptions, place medicine orders, and make online payments. Pharmacists verify prescriptions and dispense medicines. Administrators manage medicine inventory and user accounts.
Identify all actors:

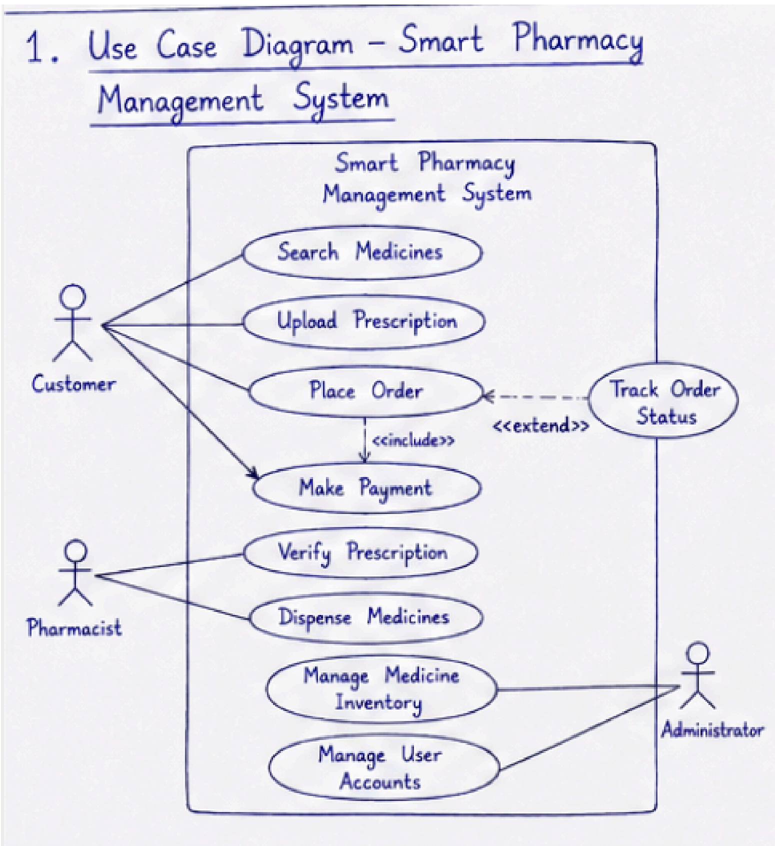
- Customer
- Pharmacist
- Administrator

Identify all use cases:

- Search Medicines
- Upload Prescription
- Place Medicine Order
- Make Online Payment
- Track Order Status
- Verify Prescription
- Dispense Medicines
- Manage Medicine Inventory
- Manage User Accounts

Design a Use Case Diagram showing appropriate relationships:

- Association between actors and use cases
- <<include>> : Place Order → Make Online Payment
- <<extend>> : Track Order Status → Place Order
- Generalization (if applicable)



2. Class Diagram Scenario

A **Smart Hotel Management System** consists of entities such as Guest, Room, Reservation, Payment, and Receptionist. The existing design does not clearly define class attributes, operations, or relationships among these entities.
Based on this scenario, identify the required classes, define suitable attributes and methods, establish relationships (**association, aggregation, composition, and inheritance**), specify multiplicity, and construct a detailed **Class Diagram**.
(5 Marks)
Answer:
Identify the required classes:

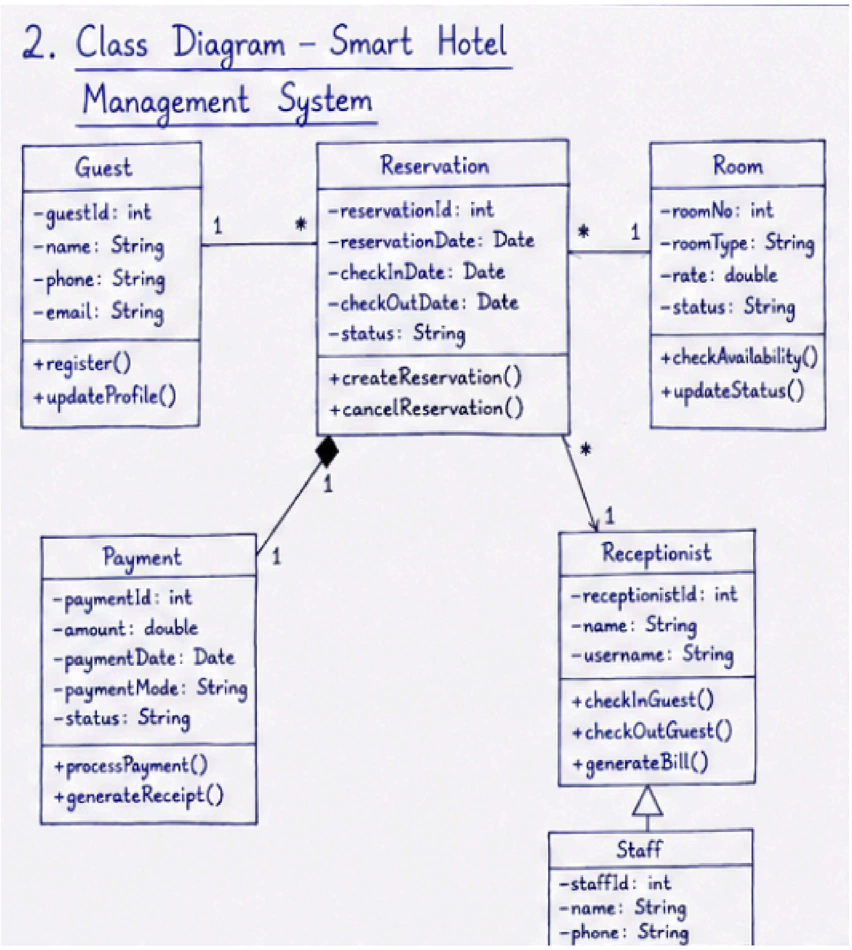
- Guest
- Room
- Reservation
- Payment
- Receptionist

Define suitable attributes and methods for each class.
Establish relationships:

- Association
- Aggregation
- Composition
- Inheritance

Specify multiplicity:

- Guest (1) → Reservation (*)
- Reservation (*) → Room (1)



3. Sequence Diagram Scenario

In an **Online Banking System**, a customer logs in, transfers funds, verifies the transaction using OTP, and receives a transaction confirmation. The communication among the customer, authentication service, banking server, and database is incomplete.
Based on this scenario, analyze the sequence of operations, identify participating objects and message flows, and design a **Sequence Diagram** representing the complete interaction process.
(5 Marks)
Answer:
Analyze the sequence of operations:

- Customer logs in.
- Customer transfers funds.
- OTP verification is performed.
- Transaction confirmation is sent.

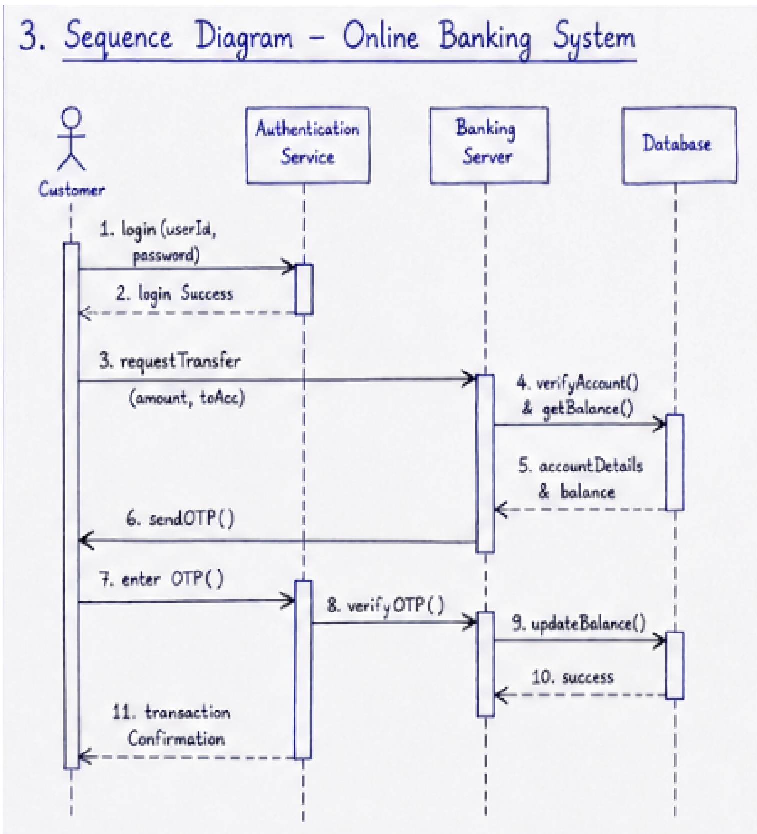
Identify participating objects:

- Customer
- Authentication Service
- Banking Server
- Database

Identify message flows:

- Login()

- Verify User
- Request Transfer
- Verify Account
- Send OTP
- Verify OTP
- Update Database
- Send Confirmation



4. Activity Diagram Scenario

A **Passport Application System** allows applicants to submit applications, upload supporting documents, complete identity verification, pay processing fees, and receive passports. The workflow does not clearly represent approval, rejection, or concurrent activities.
Based on this scenario, analyze the workflow, identify decision points and parallel activities, and design an optimized **Activity Diagram** using decision nodes, forks, and joins.
(5 Marks)
Answer:
Analyze the workflow:

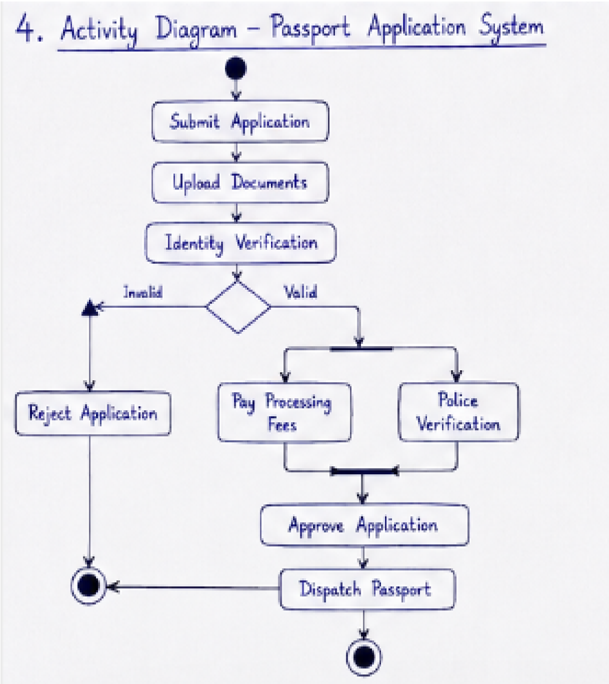
- Submit Application
- Upload Documents
- Identity Verification
- Pay Processing Fee
- Police Verification
- Approve Application
- Dispatch Passport

Identify decision points:

- Documents Valid?
- Application Approved?

Identify concurrent activities:

- Fee Payment
- Police Verification



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5. State Diagram Scenario

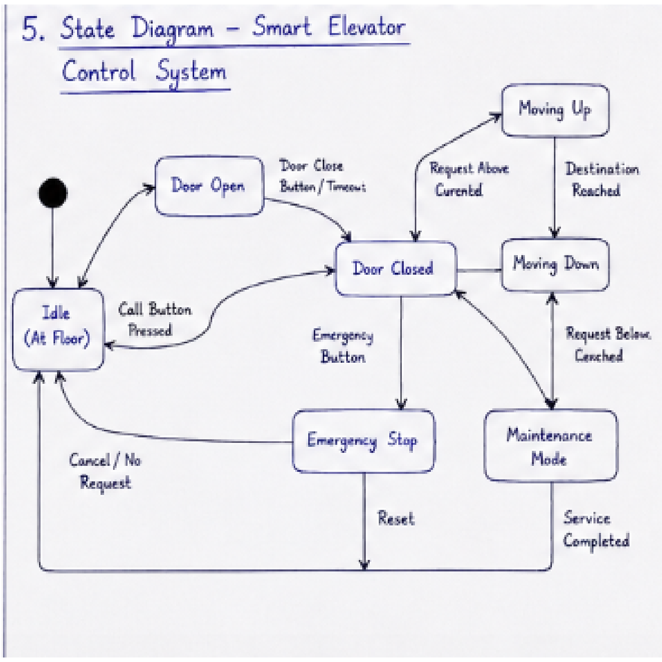
A **Smart Elevator Control System** operates through states such as Idle, Door Open, Door Closed, Moving Up, Moving Down, Emergency Stop, and Maintenance Mode. The transitions between these states are not clearly specified.
Based on this scenario, identify all possible states, triggering events, and transitions, and construct a **State Transition Diagram** that accurately represents the elevator's behavior.
(5 Marks)

Identify all possible states:

- Idle
- Door Open
- Door Closed
- Moving Up
- Moving Down
- Emergency Stop
- Maintenance Mode

Identify triggering events:

- Call Button Pressed
- Door Close Button
- Destination Reached
- Emergency Button
- Reset
- Maintenance Enabled



6. Deployment Diagram Scenario

A **Smart Home Automation System** uses smart sensors, IoT controllers, a home gateway, cloud servers, and a mobile application to monitor and control lighting, security, and appliances remotely. The deployment architecture does not clearly illustrate hardware nodes, deployed software, or communication channels.
Based on this scenario, analyze the deployment environment, identify all hardware nodes and software artifacts, and design a **Deployment Diagram** showing the physical architecture and communication links.
(5 Marks)
Answer:
Identify all hardware nodes:

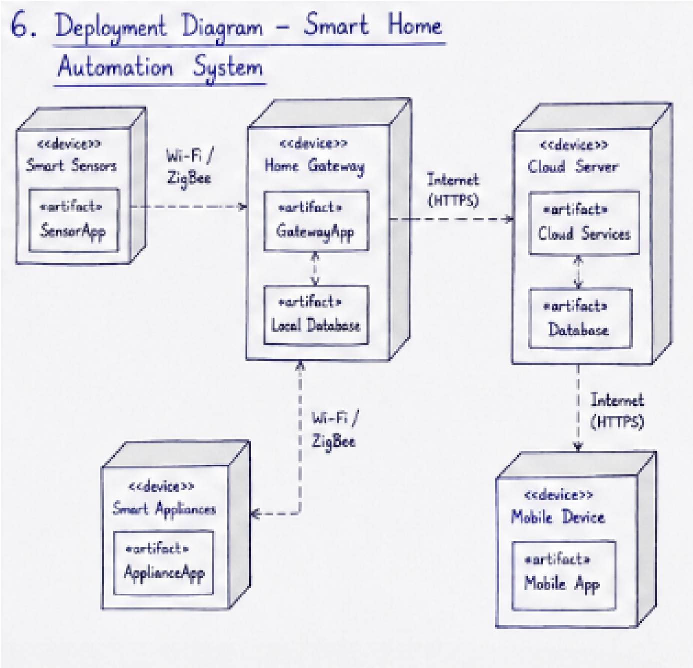
- Smart Sensors
- IoT Controller
- Home Gateway
- Cloud Server
- Mobile Device

Identify all software artifacts:

- Sensor Software
- Gateway Software
- Cloud Services
- Mobile Application

Identify communication links:

- Wi-Fi
- ZigBee
- Internet (HTTPS)



RUBRICS FOR CALCULATION

Scenario-Based

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand